Test 2

- Definitions up to end of chapter 9
- Practical up through 8.2 theorem 4
- Starts at “things not on test 1 (i.e 2.5) “
- Chapters 1 & 2 still essential.

Chapter 9 Relations

Definition

If A and B are sets, then a relation from A to B is a subset of $A \times B$.

if R is such a relation, then for any $(x, y) \in A \times B$, either

$(x, y) \in R$ (We say x is related to y and write it as xRy)

or

$(x, y) \notin R$ (x is not related to y $x \not R y$)

Examples of relations include

$=<, \leq, \rightarrow, \text{etc}$

Example

$5 < 6$ means $(5, 6) \in \{(x, y) \mid x \text{ is less than } y\}$

We actually have seen subsets of cartesian products before.

In functions, every graph is a subset of $\mathbb{R}^2$ or $R \times R$

If,

$f : A \rightarrow B$

is a function, we can build a corresponding relation:

$R = \{(x, y) \in A \times B \mid f(x) = y\}$

IN this way, every function is a relation.

But not vice versa:

$y = \pm\sqrt{x}$ is not a function

Relations can involve any domain and codomain, (in which case grahps might not be appropriate)

Example

Suppose $S = \{\text{Alice, Bob, Eve}\}$ and

$V = \{\text{Halo, Mariocart, Bash Brothers}\}$

A relation, R could track which student has played which game.

WE could use a table:

	Halo	Mariocart	Bash Brothers
Alice	x	x	
Bob		x	
Eve	x		

or sometimes an array or matrix.

Or directed graph can encod this information

Alice -> Halo, Bash Brothers

Bob -> Mariocart and Halo

Eve ->

If the domain and codomain are identical, say both are A, we say the relation is a relation <u>on</u> A

"R is a relation on A"

is shorthand for "R is a relation from A to A, hey look, A and A are the same."

Example

$/(\text{Divide})$ is a relation on $\mathbb{N}$

we say $\frac{a}{b}$ iff $\exists k(b = ak)$ (i.e $\frac{b}{a}$ is an integer)

Restricting to $A = \{2, 3, 4, 5, 6, 7, 8\}$ We have

	2	3	4	5	6	7	8
2	x		x		x		x
3		x			x		
4			x				x
5				x			
6					x		
7						x	
8							x

Same idea but with a directed graph.

Operations on Relations

RElations are just sets, so we can apply set operations like $\cap, \cup, \setminus, \complement$

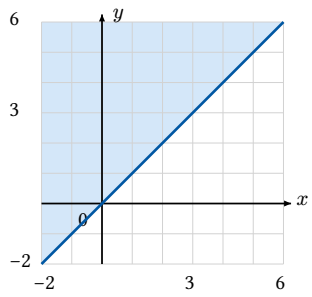
Example

$R = \{(x, y) \mid x \leq y\}$

$S = \{(x, y) \mid x \leq 3\}$

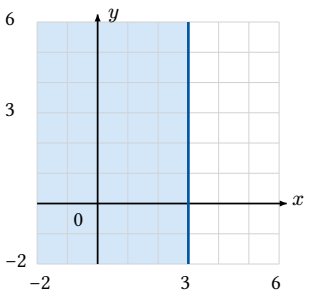
R

$x \leq y$



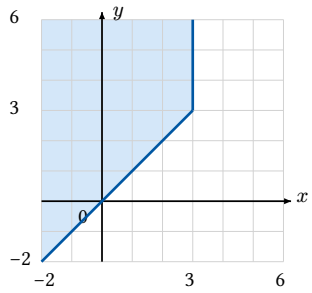
S

$x \leq 3$



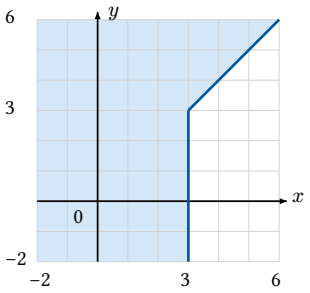
$R \cap S$

$x \leq y$ and $x \leq 3$



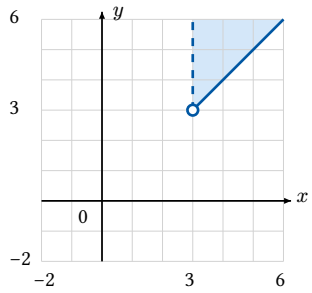
$R \cup S$

$x \leq y$ or $x \leq 3$



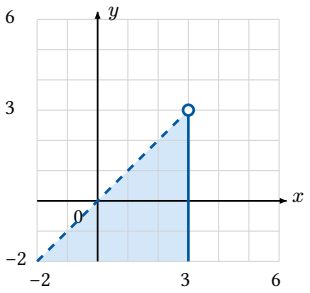
$R \setminus S$

$x \leq y$ and $x > 3$



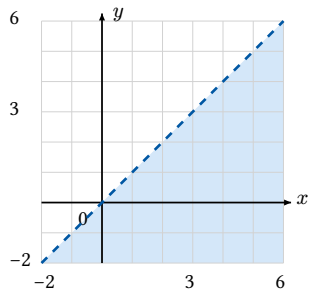
$S \setminus R$

$x \leq 3$ and $y < x$



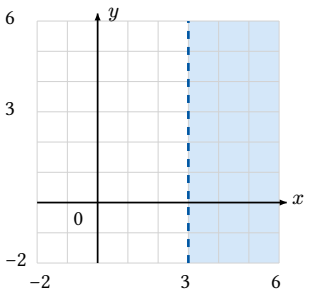
R^c

$y < x$



S^c

$x > 3$



With a bit of creative notation, $\leq$ is $< \cup =$

We can also compose relations

Example

If R relates student s to games V with xRy means x has played y

and T related to V to C via yTz means character z is in game y

We can ask:

Has x ever played a game featuring character z

This would be true iff there exists some game with two properties:

xRg and gTz

We'll write

$x(T \circ R)(z)$

This is an extension of function composition.

If $f : A \to B$ and $g : B \to C$

then $(g \circ f(x)) = z$

means $g(f(x)) = z$ (Call $f(x)$ by the name y)

so $\exists y(f(x) = y \wedge f(y) = z)$